

Module -4

Applications of Laplace Transform for circuits with initial conditions

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SENSE

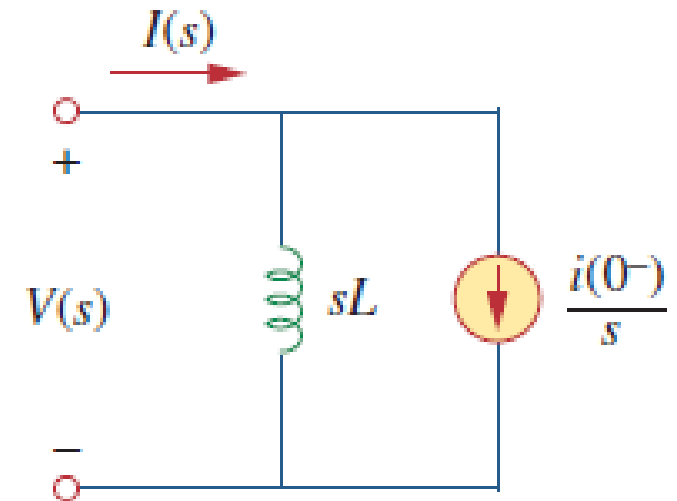
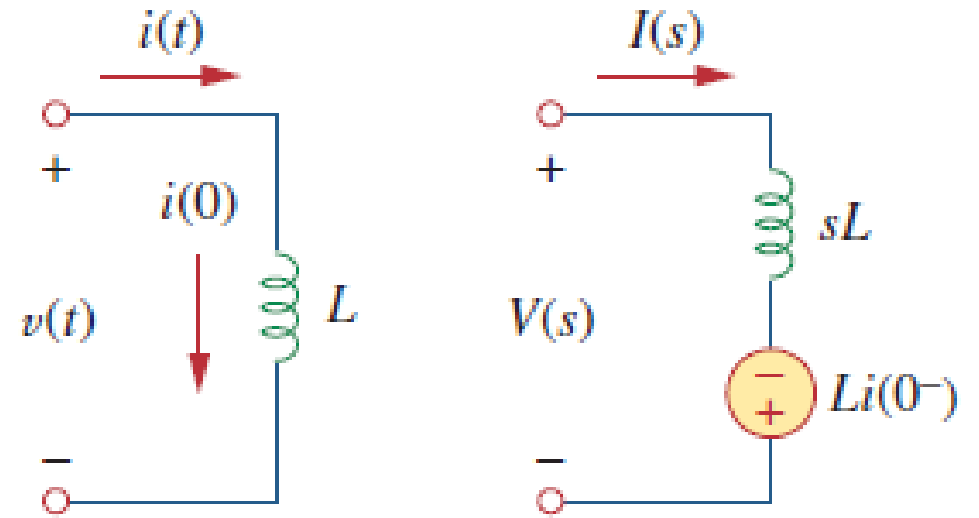
Inductor

$$v(t) = L \frac{di(t)}{dt}$$

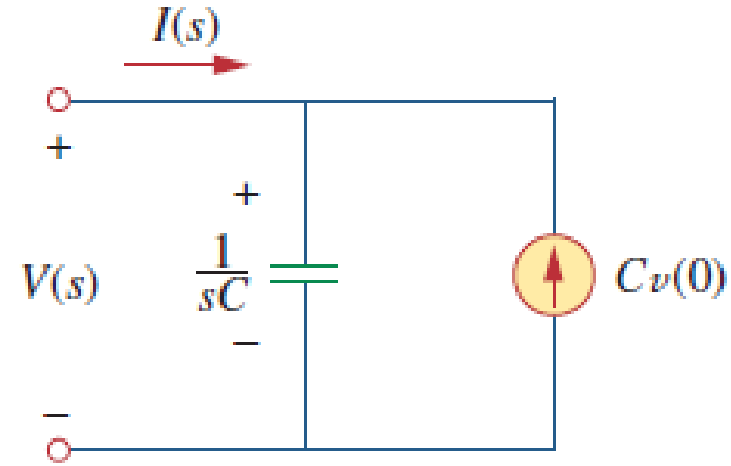
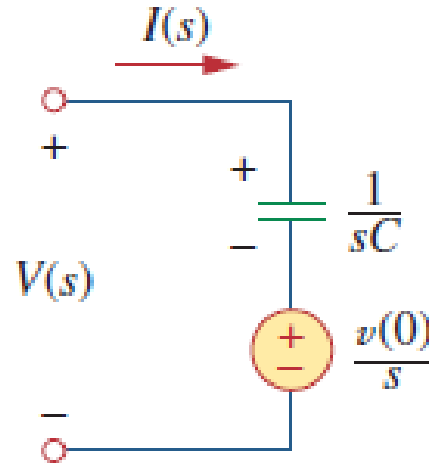
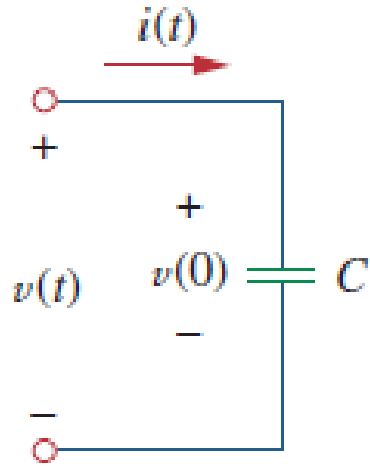
Taking the Laplace transform of both sides

$$V(s) = L[sI(s) - i(0^-)] = sLI(s) - Li(0^-)$$

$$I(s) = \frac{1}{sL} V(s) + \frac{i(0^-)}{s}$$



Capacitor



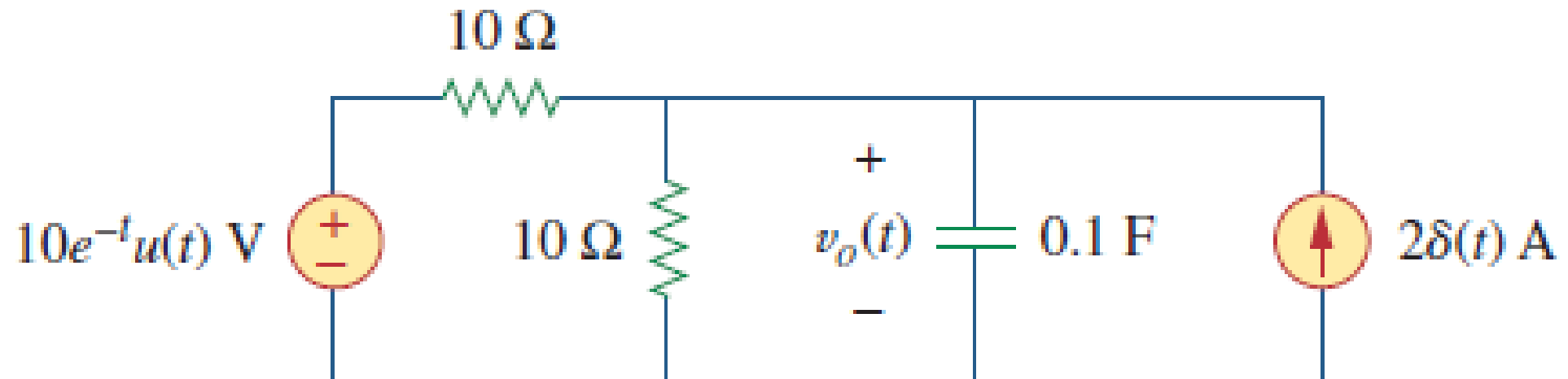
$$i(t) = C \frac{dv(t)}{dt}$$

$$I(s) = C[sV(s) - v(0^-)] = sCV(s) - Cv(0^-)$$

$$V(s) = \frac{1}{sC}I(s) + \frac{v(0^-)}{s}$$

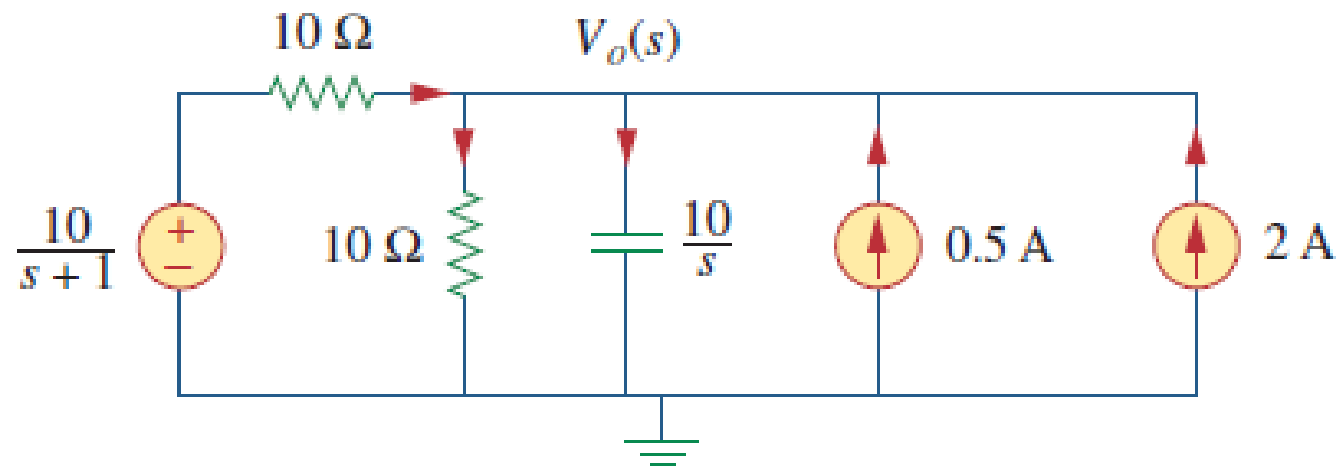
Numerical - 1

Find $v_o(t)$ in the circuit. Assume $v_o(0) = 5$ V.



Solution:

We transform the circuit to the s -domain



We apply nodal analysis. At the top node,

$$\frac{10/(s+1) - V_o}{10} + 2 + 0.5 = \frac{V_o}{10} + \frac{V_o}{10/s}$$

$$\frac{1}{s+1} + 2.5 = \frac{2V_o}{10} + \frac{sV_o}{10} = \frac{1}{10}V_o(s+2)$$

Multiplying through by 10,

$$\frac{10}{s+1} + 25 = V_o(s+2)$$

$$V_o = \frac{25s+35}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

where

$$A = (s+1)V_o(s) \Big|_{s=-1} = \frac{25s+35}{(s+2)} \Big|_{s=-1} = \frac{10}{1} = 10$$

$$B = (s+2)V_o(s) \Big|_{s=-2} = \frac{25s+35}{(s+1)} \Big|_{s=-2} = \frac{-15}{-1} = 15$$

Thus,

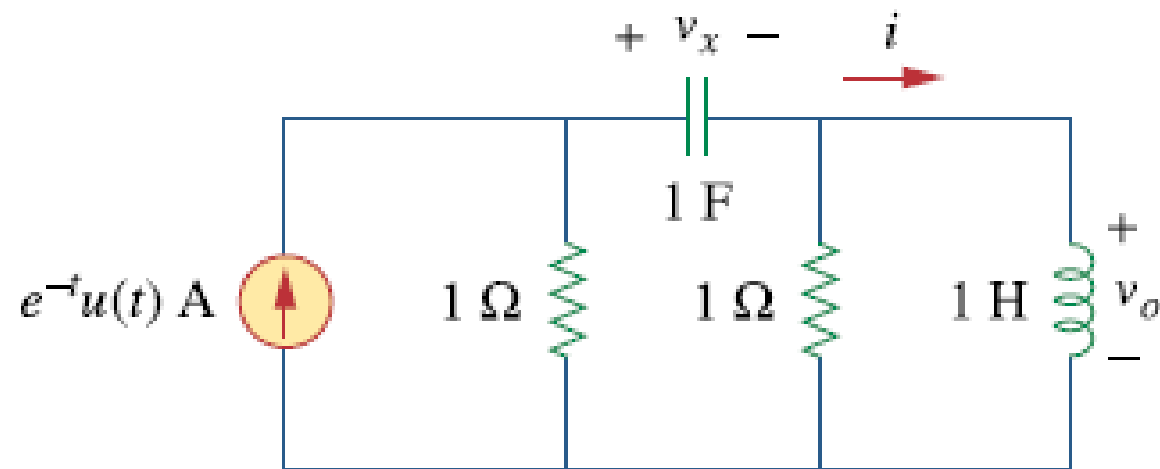
$$V_o(s) = \frac{10}{s+1} + \frac{15}{s+2}$$

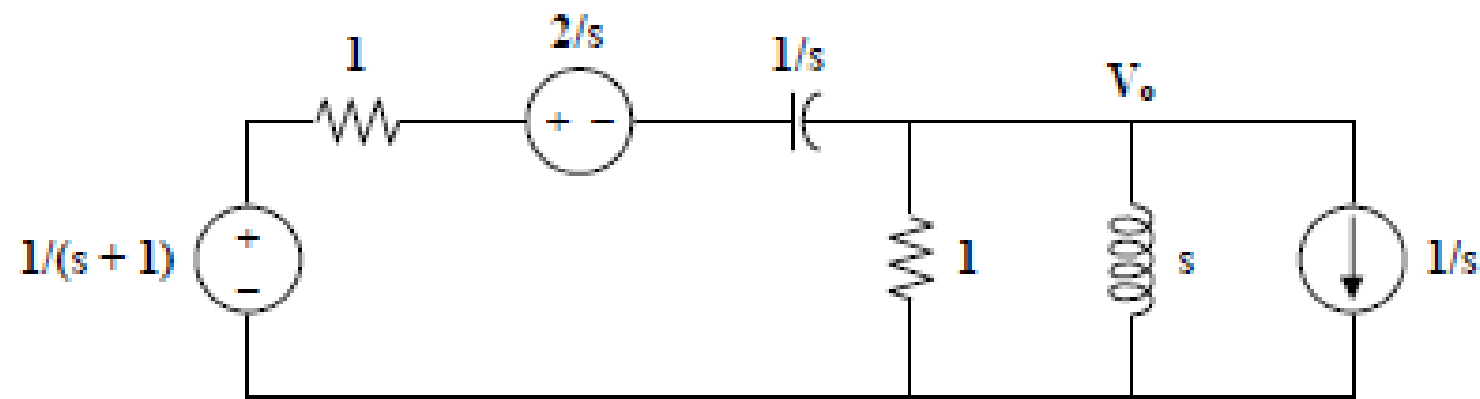
Taking the inverse Laplace transform, we obtain

$$v_o(t) = (10e^{-t} + 15e^{-2t})u(t) \text{ V}$$

Numerical - 2

Find $v_o(t)$ in the circuit if $v_x(0) = 2\text{ V}$ and $i(0) = 1\text{ A}$.





At the main non-reference node, KCL gives

$$\frac{1/(s+1) - 2/s - V_o}{1 + 1/s} = \frac{V_o}{1} + \frac{V_o}{s} + \frac{1}{s}$$

$$\frac{s}{s+1} - 2 - sV_o = (s+1)(1 + 1/s)V_o + \frac{s+1}{s}$$

$$\frac{s}{s+1} - \frac{s+1}{s} - 2 = (2s + 2 + 1/s)V_o$$

$$V_o = \frac{-2s^2 - 4s - 1}{(s+1)(2s^2 + 2s + 1)}$$

$$V_o = \frac{-s - 2s - 0.5}{(s+1)(s^2 + s + 0.5)} = \frac{A}{s+1} + \frac{Bs + C}{s^2 + s + 0.5}$$

$$A = (s+1) V_o \Big|_{s=-1} = 1$$

$$-s^2 - 2s - 0.5 = A(s^2 + s + 0.5) + B(s^2 + s) + C(s+1)$$

Equating coefficients :

$$s^2: \quad -1 = A + B \quad \longrightarrow \quad B = -2$$

$$s^1: \quad -2 = A + B + C \quad \longrightarrow \quad C = -1$$

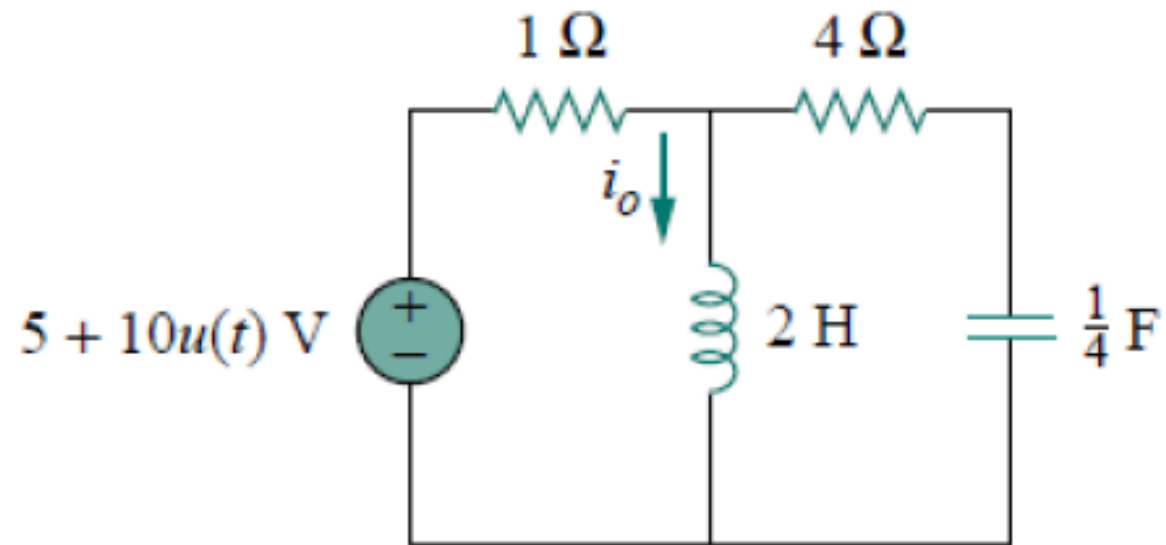
$$s^0: \quad -0.5 = 0.5A + C = 0.5 - 1 = -0.5$$

$$V_o = \frac{1}{s+1} - \frac{2s+1}{s^2+s+0.5} = \frac{1}{s+1} - \frac{2(s+0.5)}{(s+0.5)^2 + (0.5)^2}$$

$$\underline{v_o(t) = \left[e^{-t} - 2e^{-t/2} \cos(t/2) \right] u(t) \text{ V}}$$

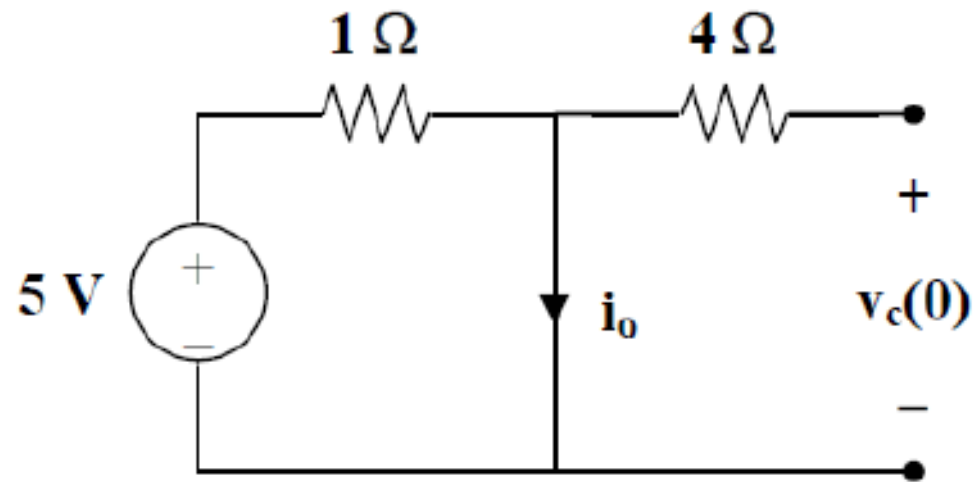
Numerical - 3

Find $i_o(t)$ in the circuit shown in figure below.



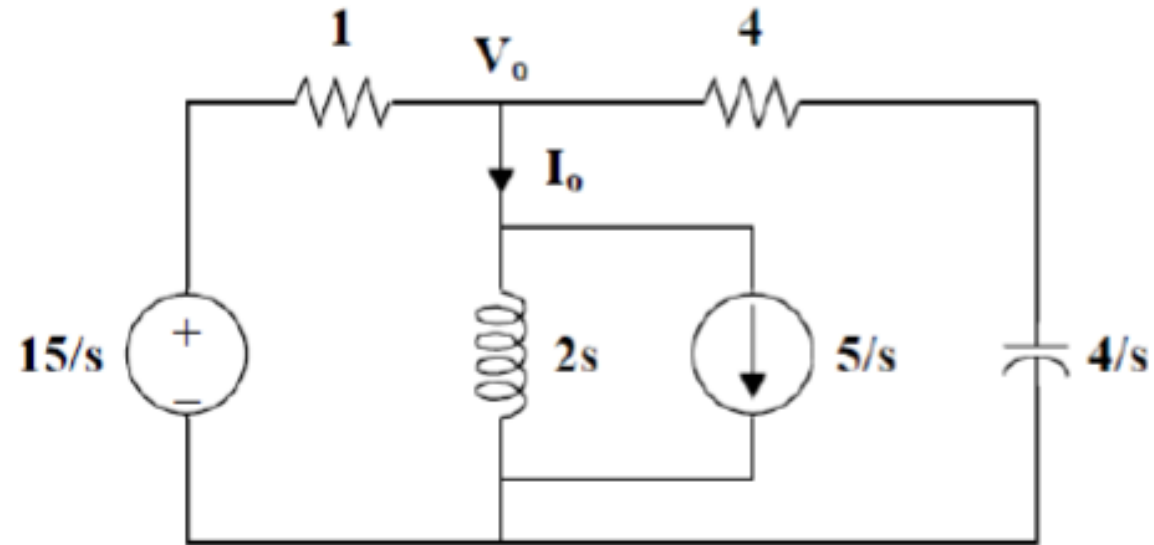
Solution:

We first find the initial conditions from the circuit



$$i_o(0^-) = 5 \text{ A}, \quad v_c(0^-) = 0 \text{ V}$$

S domain circuit:



At node o,

$$\frac{V_o - 15/s}{1} + \frac{V_o}{2s} + \frac{5}{s} + \frac{V_o - 0}{4 + 4/s} = 0$$

$$\frac{15}{s} - \frac{5}{s} = \left(1 + \frac{1}{2s} + \frac{s}{4(s+1)} \right) V_o$$

$$\frac{10}{s} = \frac{4s^2 + 4s + 2s + 2 + s^2}{4s(s+1)} V_o = \frac{5s^2 + 6s + 2}{4s(s+1)} V_o$$

$$V_o = \frac{40(s+1)}{5s^2 + 6s + 2}$$

$$I_o = \frac{V_o}{2s} + \frac{5}{s} = \frac{4(s+1)}{s(s^2 + 1.2s + 0.4)} + \frac{5}{s}$$

$$I_o = \frac{5}{s} + \frac{A}{s} + \frac{Bs + C}{s^2 + 1.2s + 0.4}$$

$$4(s+1) = A(s^2 + 1.2s + 0.4) + Bs + Cs$$

Equating coefficients :

$$s^0: \quad 4 = 0.4A \quad \longrightarrow \quad A = 10$$

$$s^1: \quad 4 = 1.2A + C \quad \longrightarrow \quad C = -1.2A + 4 = -8$$

$$s^2: \quad 0 = A + B \quad \longrightarrow \quad B = -A = -10$$

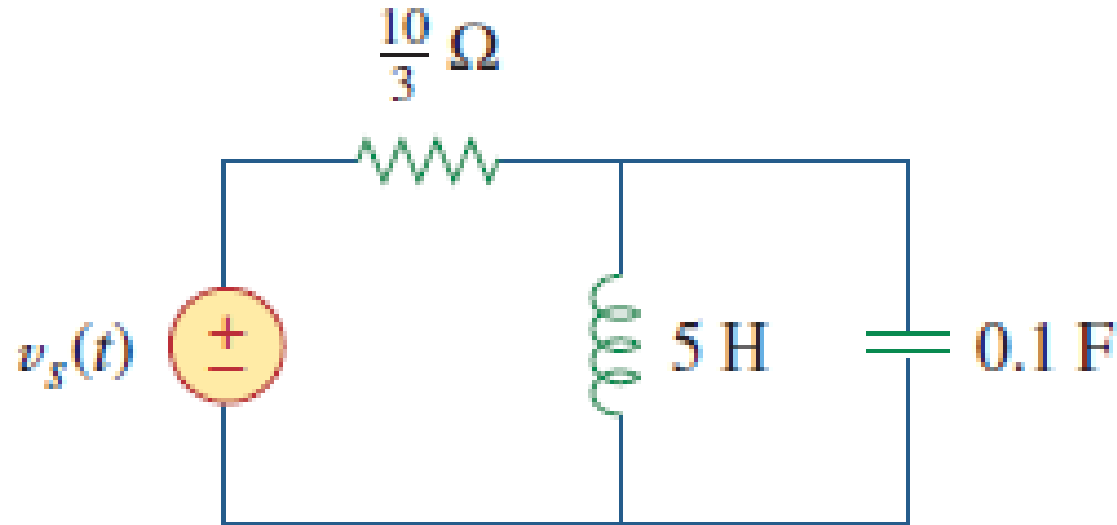
$$I_o = \frac{5}{s} + \frac{10}{s} - \frac{10s + 8}{s^2 + 1.2s + 0.4}$$

$$I_o = \frac{15}{s} - \frac{10(s + 0.6)}{(s + 0.6)^2 + 0.2^2} - \frac{10(0.2)}{(s + 0.6)^2 + 0.2^2}$$

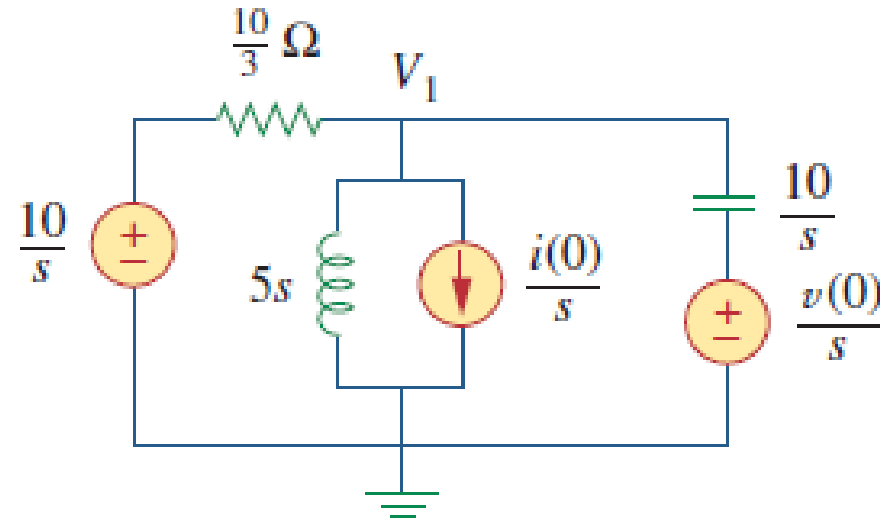
$$i_o(t) = \underline{\underline{\left[15 - 10 e^{-0.6t} (\cos(0.2t) - \sin(0.2t)) \right] u(t) \text{ A}}}$$

Numerical - 4

- Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10u(t)$ and assume that at $t = 0$, -1A flows through the inductor and $+5\text{ V}$ is across the capacitor.



Solution:



$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + \frac{i(0)}{s} + \frac{V_1 - [v(0)/s]}{1/(0.1s)} = 0$$

$$0.1 \left(s + 3 + \frac{2}{s} \right) V_1 = \frac{3}{s} + \frac{1}{s} + 0.5$$

where $v(0) = 5$ V and $i(0) = -1$ A. Simplifying we get

$$(s^2 + 3s + 2) V_1 = 40 + 5s$$

$$V_1 = \frac{40 + 5s}{(s + 1)(s + 2)} = \frac{35}{s + 1} - \frac{30}{s + 2}$$

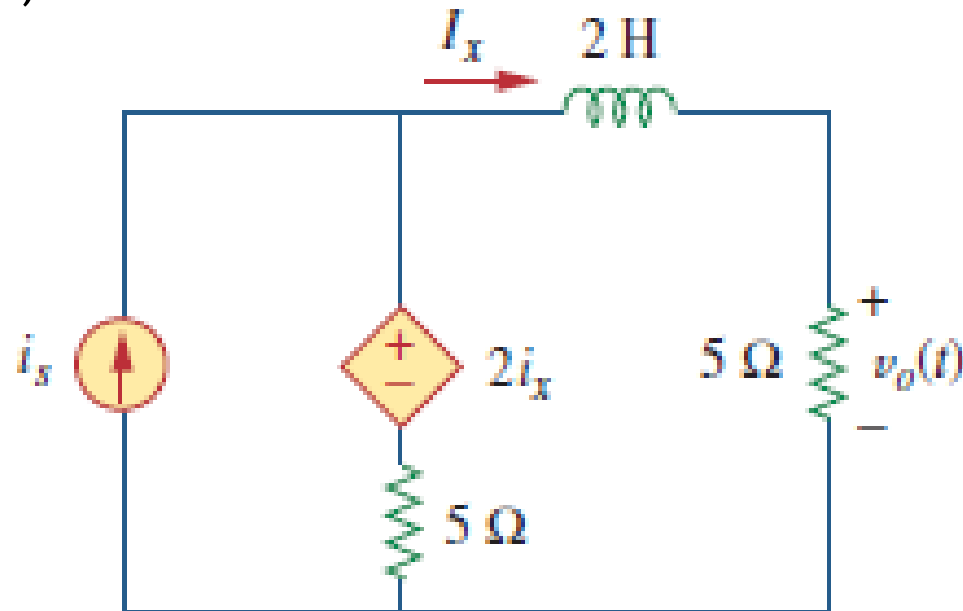
Taking the inverse Laplace transform yields

$$v_1(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V}$$

Numerical - 5

Assume that there is no initial energy stored in the circuit at $t = 0$ and that $i_s = 10u(t)$ A.

- (a) Find $V_o(s)$ using Thevenin's theorem.
- (b) Apply the initial and final value theorems to find $v_o(0+)$ and $v_o(\infty)$.
- (c) Determine $v_o(t)$.

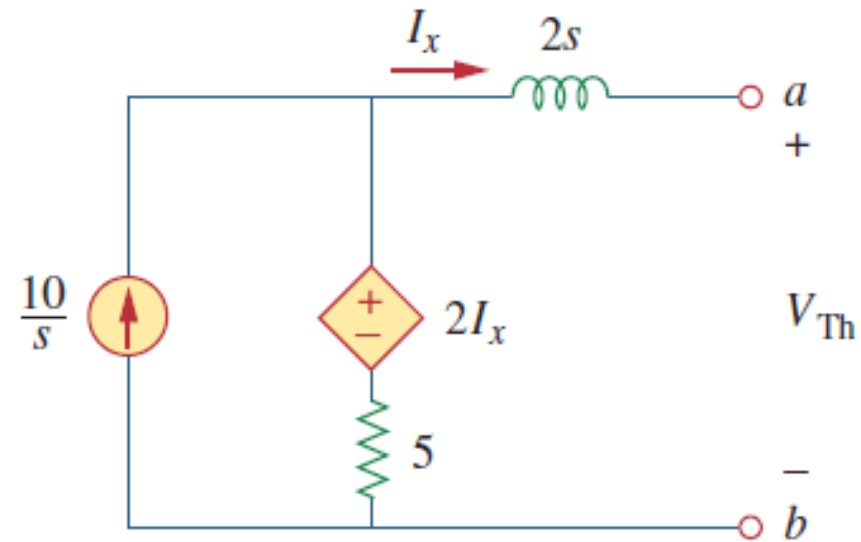


Solution:

Since there is no initial energy stored in the circuit, we assume that the initial inductor current and initial capacitor voltage are zero at $t = 0$.

(a) To find the Thevenin equivalent circuit

$$V_{oc} = V_{Th} = 5\left(\frac{10}{s}\right) = \frac{50}{s}$$



To find Z_{Th} , ($I_{\text{sc}} = I_x = V_1/2s$).

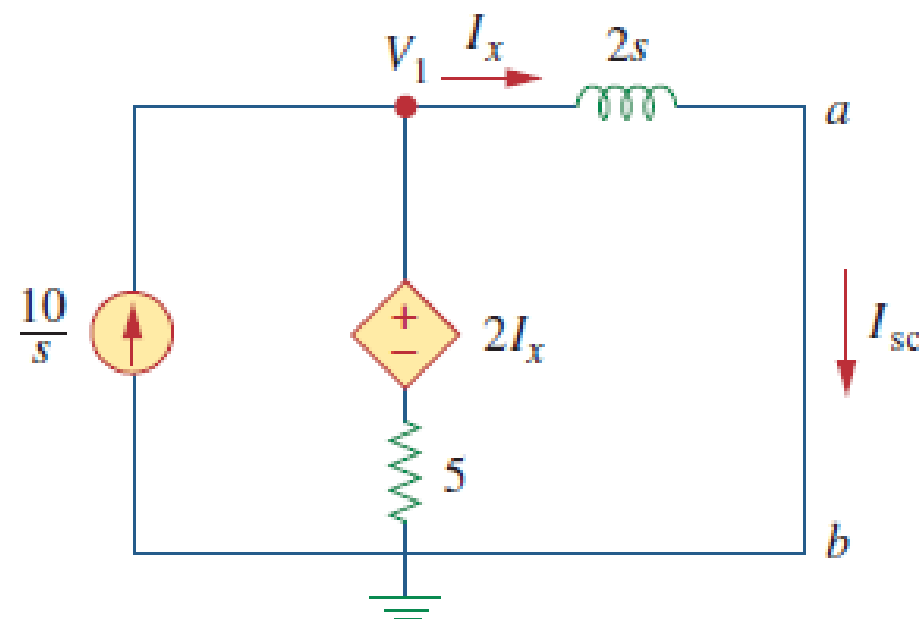
$$-\frac{10}{s} + \frac{(V_1 - 2I_x) - 0}{5} + \frac{V_1 - 0}{2s} = 0$$

$$I_x = \frac{V_1}{2s}$$

$$V_1 = \frac{100}{2s + 3}$$

$$I_{\text{sc}} = \frac{V_1}{2s} = \frac{100/(2s + 3)}{2s} = \frac{50}{s(2s + 3)}$$

$$Z_{\text{Th}} = \frac{V_{\text{oc}}}{I_{\text{sc}}} = \frac{50/s}{50/[s(2s + 3)]} = 2s + 3$$



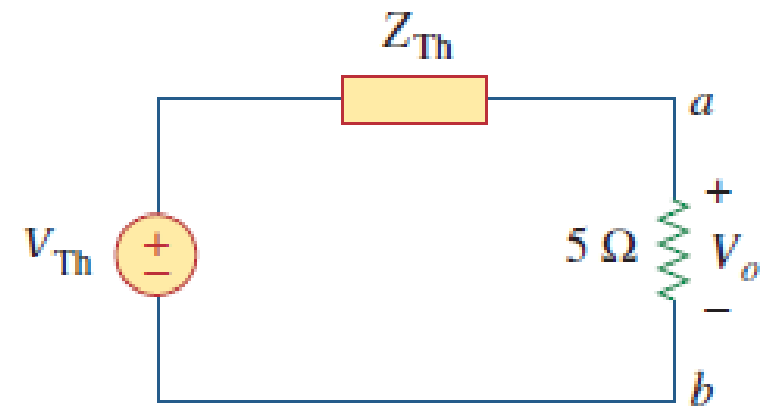
$$V_o = \frac{5}{5 + Z_{Th}} V_{Th} = \frac{5}{5 + 2s + 3} \left(\frac{50}{s} \right) = \frac{250}{s(2s + 8)} = \frac{125}{s(s + 4)}$$

(b) Using the initial-value theorem we find

$$v_o(0) = \lim_{s \rightarrow \infty} sV_o(s) = \lim_{s \rightarrow \infty} \frac{125}{s + 4} = \lim_{s \rightarrow \infty} \frac{125/s}{1 + 4/s} = \frac{0}{1} = 0$$

Using the final-value theorem we find

$$v_o(\infty) = \lim_{s \rightarrow 0} sV_o(s) = \lim_{s \rightarrow 0} \frac{125}{s + 4} = \frac{125}{4} = 31.25 \text{ V}$$



(c) By partial fraction,

$$V_o = \frac{125}{s(s + 4)} = \frac{A}{s} + \frac{B}{s + 4}$$

$$A = sV_o(s) \Big|_{s=0} = \frac{125}{s + 4} \Big|_{s=0} = 31.25$$

$$B = (s + 4)V_o(s) \Big|_{s=-4} = \frac{125}{s} \Big|_{s=-4} = -31.25$$

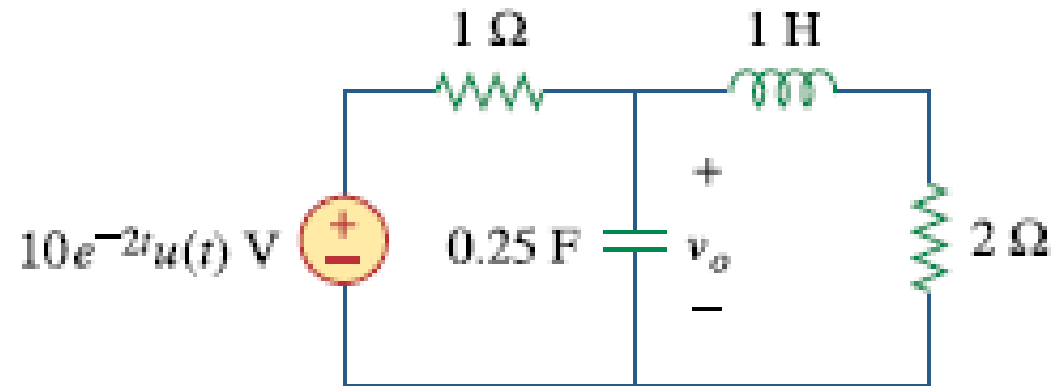
$$V_o = \frac{31.25}{s} - \frac{31.25}{s + 4}$$

Taking the inverse Laplace transform gives

$$v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$$

Numerical - 6

Use Thevenin's theorem to determine $v_o(t)$, $t > 0$ in the circuit

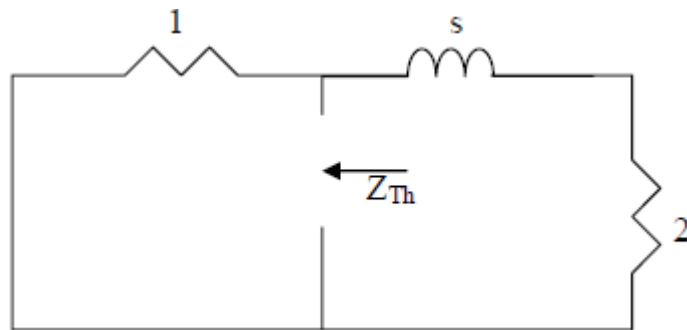


Solution

$1H \longrightarrow 1s$ and $i_L(0) = 0$ (the sources is zero for all $t < 0$).

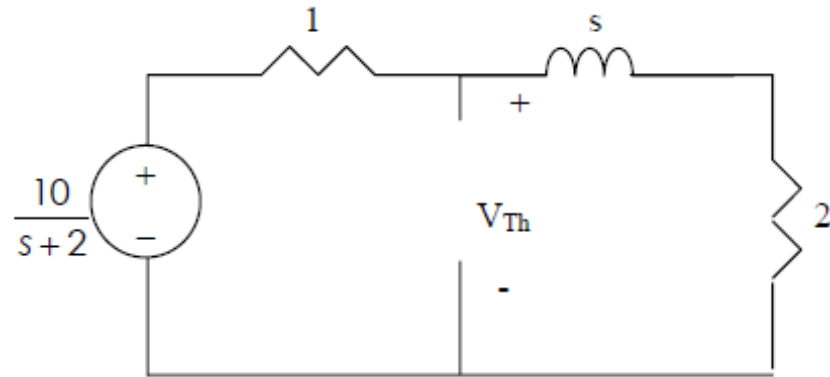
$\frac{1}{4}F \longrightarrow \frac{1}{sC} = \frac{4}{s}$ and $v_C(0) = 0$ (again, there are no source contributions for all $t < 0$).

To find Z_{Th} , consider the circuit below.



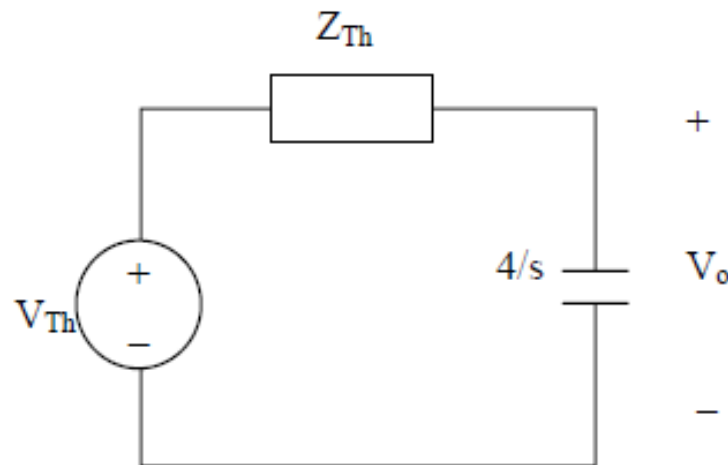
$$Z_{Th} = 1 // (s + 2) = \frac{s + 2}{s + 3}$$

To find V_{Th} , consider the circuit below.



$$V_{Th} = \frac{s+2}{s+3} \square \frac{10}{s+2} = \frac{10}{s+3}$$

The Thevenin equivalent circuit is shown below



$$V_o = \frac{\frac{4}{s}}{\frac{4}{s} + Z_{Th}} V_{Th} = \frac{\frac{4}{s}}{\frac{4}{s} + \frac{s+2}{s+3}} \square \frac{10}{s+3} = \frac{40}{s^2 + 6s + 12} = \frac{\frac{40}{\sqrt{3}} \sqrt{3}}{(s+3)^2 + (\sqrt{3})^2}$$

$$v_o(t) = \underline{23.094 e^{-3t} \sin \sqrt{3} t}$$

Numerical - 6

Apply the Thevenin's theorem to determine $v_o(t)$ for $t > 0$ in the circuit of Fig.(c) using Laplace transform.

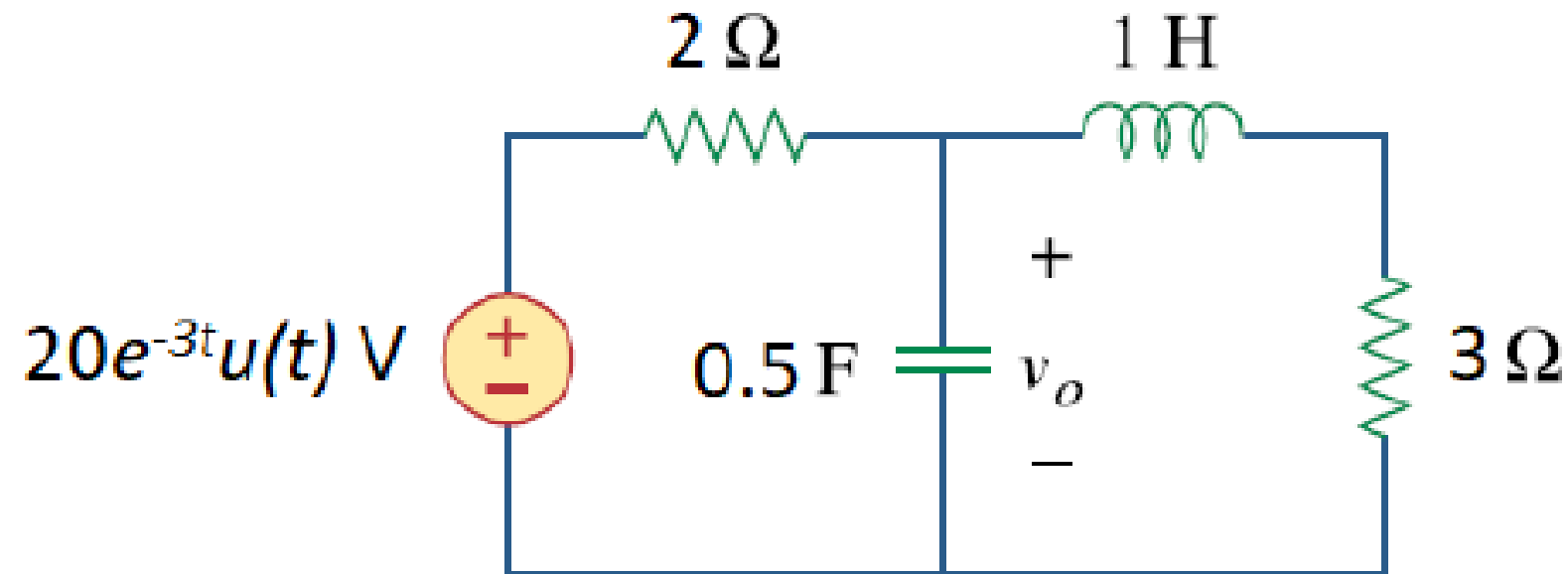


Fig. (c)